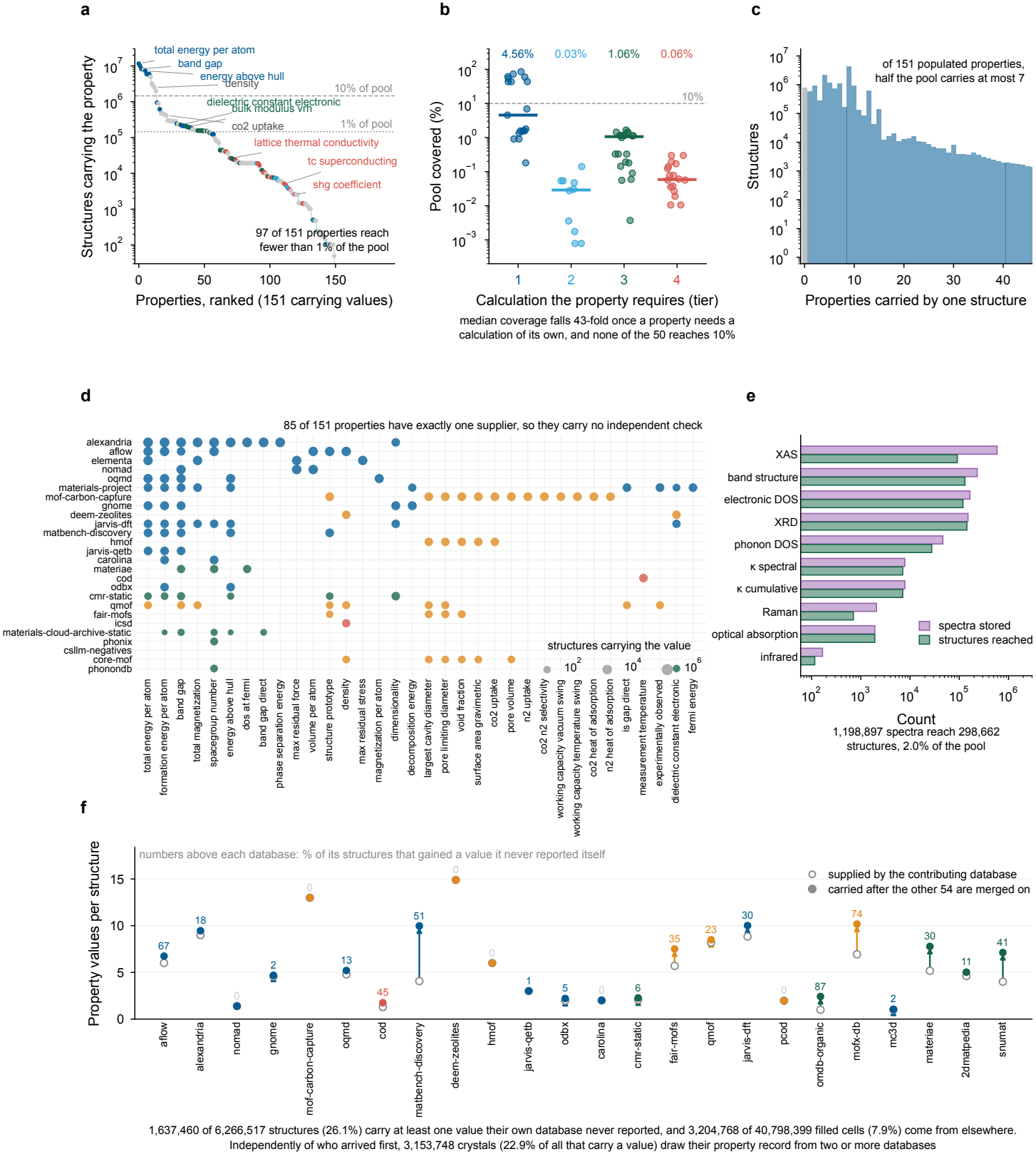

a
Structures carrying the property
total energy per atom
band gap
energy above hull
density
10% of pool
dielectric constant electronic
bulk modulus vrh
co2 uptake
1% of pool
lattice thermal conductivity
tc superconducting
shg coefficient
97 of 151 properties reach fewer than 1% of the pool
Properties, ranked (151 carrying values)
b
4.56%
0.03%
1.06%
0.06%
10%
Pool covered (%)
Calculation the property requires (tier)
median coverage falls 43-fold once a property needs a calculation of its own, and none of the 50 reaches 10%
c
of 151 populated properties, half the pool carries at most 7
Structures
Properties carried by one structure
d
85 of 151 properties have exactly one supplier, so they carry no independent check
alexandria
aflow
elementa
nomad
oqmd
materials-project
mof-carbon-capture
gnome
deem-zeolites
jarvis-dft
matbench-discovery
hmof
jarvis-qetb
carolina
materiae
cod
odbx
cmr-static
qmof
fair-mofs
icsd
materials-cloud-archive-static
phonix
csllm-negatives
core-mof
phonondb
structures carrying the value
total energy per atom
formation energy per atom
band gap
total magnetization
spacegroup number
energy above hull
dos at fermi
band gap direct
phase separation energy
max residual force
volume per atom
structure prototype
density
max residual stress
magnetization per atom
dimensionality
decomposition energy
largest cavity diameter
pore limiting diameter
void fraction
surface area gravimetric
co2 uptake
pore volume
n2 uptake
co2 n2 selectivity
working capacity vacuum swing
working capacity temperature swing
co2 heat of adsorption
n2 heat of adsorption
is gap direct
measurement temperature
experimentally observed
dielectric constant electronic
fermi energy
e
XAS
band structure
electronic DOS
XRD
phonon DOS
κ spectral
κ cumulative
Raman
optical absorption
infrared
spectra stored
structures reached
Count
1,198,897 spectra reach 298,662 structures, 2.0% of the pool
f
numbers above each database: % of its structures that gained a value it never reported itself
supplied by the contributing database
carried after the other 54 are merged on
Property values per structure
aflow
alexandria
nomad
gnome
mof-carbon-capture
oqmd
cod
matbench-discovery
deem-zeolites
hmof
jarvis-qetb
odbx
carolina
cmr-static
fair-mofs
qmof
jarvis-dft
pcod
omdb-organic
mofx-db
mc3d
materiae
2dmatpedia
snumat
1,637,460 of 6,266,517 structures (26.1%) carry at least one value their own database never reported, and 3,204,768 of 40,798,399 filled cells (7.9%) come from elsewhere.
Independently of who arrived first, 3,153,748 crystals (22.9% of all that carry a value) draw their property record from two or more databases